PROBLEM NS7

Liquid Film on an Inclined Plane

Original statement · NS7.pdf

Free-surface film | Gravity-driven flow | Viscous dissipation | Energy balance

OFFICIAL PROBLEM STATEMENT

Consider the steady planar motion of a liquid of density ρ, viscosity μ, thermal conductivity k, and specific heat c that flows down over an inclined plane of angle α due to the action of gravity, forming a layer of constant thickness h. To solve the problem, use the coordinate system (x, y) indicated in the figure, in which gravity can be expressed as $\bar{g} = g\,\mathrm{sin}(\alpha)\bar{e}_x - g\,\mathrm{cos}(\alpha)\bar{e}_y$. To analyze the resulting liquid flow, assume that the velocity components $v_x(y)$ and $v_y(y)$ are a function of the distance to the wall, and that the air surrounding the liquid has a density $\rho_a \ll \rho$ and a viscosity $\mu_a \ll \mu$, such that in the first approximation it moves without significant pressure differences, exerting on the liquid a negligible viscous force.

▪ Using continuity, determine the velocity component $v_y(y)$.

Obtain the pressure field $p(x, y)$ by integration of the y component of the momentum equation, demonstrating in particular that $\partial p/\partial x = 0$.

Compute $v_x(y)$.

Calculate the force per unit surface that the liquid exerts on the wall.

Determine the temperature in the liquid $T(y)$. In the calculation, assume that the thermal conductivity of air is very small compared with that of the liquid ($k_a \ll k$) and that the wall is at constant uniform temperature T_w.

Obtain the conductive heat flux from the liquid to the wall $(\bar{q} \cdot \bar{n})_w$, where $\bar{n}$ is appropriately oriented.

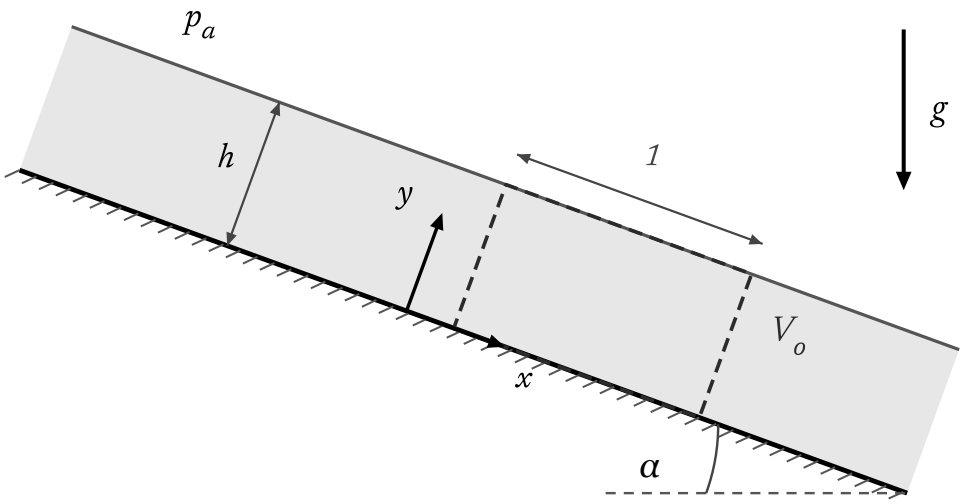
To analyze the energy balances emerging in the problem, consider the fixed control volume V_o of unity length indicated in the figure.

▪ Compute the total energy dissipated per unit time inside the control volume $\int_{V_o} \phi_v \, dV$.

▪ Calculate the work per unit time done by gravity $\int_{V_o} \rho \bar{v} \cdot \bar{g} \, dV$.

▪ Use convervation of energy in the control volume to explain why $(\bar{q} \cdot \bar{n})_w = \int_{V_o} \phi_v \, dV = \int_{V_o} \rho \bar{v} \cdot \bar{g} \, dV$.

Transcribed as printed, including the markers of the sheet and the spelling "convervation".



Statement figure (redrawn from NS7.pdf, p. 1): layer of thickness h on a plane inclined at α, axes x (down the plane) and y (normal to it), gravity $\bar{g} = g\sin\alpha\,\bar{e}_x - g\cos\alpha\,\bar{e}_y$, air at p_a and the fixed control volume V_o of unit length

Source: NS7.pdf (page 1). Notes.pdf, Chapter 4: continuity Eq. 4.40, momentum Eq. 4.41, energy Eq. 4.42, internal-energy balance Eq. 4.25, total-energy balance Eq. 4.3.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady planar flow of a perfect liquid (constant ρ, μ, k, c): equations (4.40)-(4.42).
- The fields depend only on the distance y to the plane: $\partial/\partial x = 0$ for v_x, v_y and T. The pressure is determined below.
- The plane is impermeable and fixed; the liquid layer has uniform thickness h; no chemical reaction and no radiation.
- The air exerts no pressure variation and a negligible viscous force on the liquid, and its thermal conductivity is negligible: at the free surface $y = h, p = p_a, \tau'_{xy} = 0$ and $\partial T/\partial y = 0$.

- Data: $[\rho,\ \mu,\ k,\ c,\ h,\ \alpha,\ g,\ p_a,\ T_w]$. Unknowns: $v_y, p(x,y), v_x(y)$, the wall force, $T(y)$, the wall heat flux and the three energy integrals.

Symbol	Meaning	SI unit
h	layer thickness	m
α	inclination of the plane	rad
$g\sin\alpha,\ g\cos\alpha$	components of gravity along and normal to the plane	m/s^2
p_a, T_w	air pressure, wall temperature	Pa, K
ρ, μ, k, c	liquid properties	kg/m^3, Pa s, W/(m K), J/(kg K)

PHASE 2

Frames, Field Form and Boundary Conditions

- Cartesian frame (x, y, z) attached to the plane: x down the slope, y normal to the plane and into the liquid, z out of the plane; the basis is orthonormal and right-handed. Gravity has the components $g_x = g\sin\alpha$, $g_y = -g\cos\alpha$. The frame is rotated by α with respect to the horizontal-vertical frame; the rotation matrix between them is orthogonal with determinant 1, since $\bar{e}_x = (\cos\alpha, -\sin\alpha)$ and $\bar{e}_y = (\sin\alpha, \cos\alpha)$ in horizontal-vertical components.
- Field form: $\bar{v} = v_x(y)\,\bar{e}_x + v_y(y)\,\bar{e}_y, T = T(y), p = p(x,y)$.
- Fixed control volume V_o of the figure: $0 < y < h$, unit length in x and unit width; surfaces: the wall Σ_w ($\bar{n} = -\bar{e}_y$), the free surface Σ_s ($\bar{n} = +\bar{e}_y$) and the two faces normal to x.

Location	Condition	Origin
$y = 0$	$v_x = 0, v_y = 0, T = T_w$	fixed impermeable wall at T_w
$y = h$	$p = p_a$	no pressure variations in the air
$y = h$	$\tau'_{xy} = \mu\, dv_x/dy = 0$	negligible viscous force of the air
$y = h$	$dT/dy = 0$	$k_a \ll k$: adiabatic free surface

PHASE 3

Step-by-Step Derivation

3.1 Continuity

$$\frac{\partial v_x}{\partial x} + \frac{dv_y}{dy} = \underbrace{\frac{\partial v_x}{\partial x}}_{v_x = v_x(y)} + \frac{dv_y}{dy} = 0 \implies v_y = \text{const} \tag{1}$$

The wall is impermeable, $v_y(0) = 0$, so

$$\boxed{v_y(y) = 0} \tag{2}$$

3.2 y-momentum and the pressure field

With $v_y = 0$ and $\partial_x = 0$ for the velocity, the inertial and viscous terms of the y component vanish:

$$0 = -\frac{\partial p}{\partial y} - \rho g \cos\alpha \implies p = -\rho g \cos\alpha \, y + f(x) \tag{3}$$

The free-surface condition $p(x, h) = p_a$ for every x gives $f(x) = p_a + \rho g h \cos\alpha$, a constant:

$$\boxed{p(x, y) = p_a + \rho g \cos\alpha \, (h - y)}, \qquad \frac{\partial p}{\partial x} = 0 \tag{4}$$

3.3 x-momentum and the velocity profile

$$\rho\Big(\underbrace{v_x\cancel{\frac{\partial v_x}{\partial x}}}_{\partial_x=0}+\underbrace{v_y\cancel{\frac{\partial v_x}{\partial y}}}_{v_y=0}\Big)=-\underbrace{\frac{\partial p}{\partial x}}_{=0}+\mu\frac{d^2v_x}{dy^2}+\rho g\sin\alpha \;\Longrightarrow\; \frac{d^2v_x}{dy^2}=-\frac{\rho g\sin\alpha}{\mu} \tag{5}$$

Integrating twice, $v_x=-\dfrac{\rho g\sin\alpha}{2\mu}y^2+C_1y+C_2$. The wall condition $v_x(0)=0$ gives $C_2=0$. The free-surface condition $dv_x/dy\big|_h=-\dfrac{\rho g\sin\alpha}{\mu}h+C_1=0$ gives $C_1=\dfrac{\rho g\sin\alpha}{\mu}h$:

$$\boxed{v_x(y)=\frac{\rho g\sin\alpha}{\mu}\left(hy-\frac{y^2}{2}\right)=\frac{gh^2\sin\alpha}{\nu}\left[\frac{y}{h}-\frac{1}{2}\left(\frac{y}{h}\right)^2\right]} \tag{6}$$

3.4 Force per unit surface on the wall

The normal pointing into the liquid is $\bar n=+\bar e_y$, so $(-p\bar{\bar I}+\bar{\bar\tau}')\cdot\bar n$ is the force per unit area that the liquid exerts on the wall. At $y=0$, $p=p_a+\rho gh\cos\alpha$ from (4), $\tau'_{xy}=\mu\,dv_x/dy\big|_0=\rho gh\sin\alpha$ and $\tau'_{yy}=0$:

$$\boxed{\bar f_w=-\left(p_a+\rho gh\cos\alpha\right)\bar e_y+\rho gh\sin\alpha\,\bar e_x} \tag{7}$$

The normal component pushes on the wall with the air pressure plus the hydrostatic pressure of the layer; the tangential component is the shear that drags the wall down the slope.

3.5 Temperature distribution

With $v_y=0$ and $\partial_x=0$ the energy equation (4.42) reduces to $0=\phi_v+k\,d^2T/dy^2$, with $\phi_v=\mu\,(dv_x/dy)^2$ and, from (6), $dv_x/dy=(\rho g\sin\alpha/\mu)(h-y)$:

$$\phi_v=\frac{(\rho g\sin\alpha)^2}{\mu}(h-y)^2,\qquad \frac{d^2T}{dy^2}=-\frac{(\rho g\sin\alpha)^2}{\mu k}(h-y)^2 \tag{8}$$

Integrate once: the antiderivative of $(h-y)^2$ is $-(h-y)^3/3$, so $dT/dy = \frac{(\rho g \sin\alpha)^2}{3\mu k}(h-y)^3 + B_1$. The adiabatic condition $dT/dy\big|_h = 0$ gives $B_1 = 0$. Integrating again, $T = -\frac{(\rho g \sin\alpha)^2}{12\mu k}(h-y)^4 + B_2$, and $T(0) = T_w$ gives $B_2 = T_w + \frac{(\rho g \sin\alpha)^2 h^4}{12\mu k}$:

$$\boxed{T(y) = T_w + \frac{(\rho g \sin\alpha)^2}{12\,\mu k}\left[h^4 - (h-y)^4\right]} \tag{9}$$

Expanding, $h^4 - (h-y)^4 = y\left(4h^3 - 6h^2 y + 4hy^2 - y^3\right)$.

3.6 Heat flux to the wall

The heat flux from the liquid to the wall is $\bar{q}\cdot\bar{n}$ with $\bar{n} = -\bar{e}_y$ (out of the liquid) and $\bar{q} = -k\nabla T$:

$$(\bar{q}\cdot\bar{n})_w = k\frac{dT}{dy}\Big|_{y=0} = k\,\frac{(\rho g\sin\alpha)^2 h^3}{3\mu k} = \boxed{\frac{(\rho g\sin\alpha)^2 h^3}{3\mu}} \tag{10}$$

3.7 Energy balances in the fixed control volume V_o

Total viscous dissipation. The volume V_o has unit length and width and height h, so $dV = dy$:

$$\int_{V_o}\phi_v\,dV = \int_0^h \frac{(\rho g\sin\alpha)^2}{\mu}(h-y)^2\,dy = \frac{(\rho g\sin\alpha)^2}{\mu}\left[-\frac{(h-y)^3}{3}\right]_0^h = \frac{(\rho g\sin\alpha)^2 h^3}{3\mu} \tag{11}$$

Work of gravity. $\bar{v}\cdot\bar{g} = v_x g\sin\alpha$ since $v_y = 0$. The antiderivative of $hy - y^2/2$ is $hy^2/2 - y^3/6$ and Barrow's rule gives $h^3/2 - h^3/6 = h^3/3$:

$$\int_{V_o}\rho\bar{v}\cdot\bar{g}\,dV = \rho g\sin\alpha\int_0^h v_x\,dy = \frac{\rho^2 g^2\sin^2\alpha}{\mu}\left[\frac{hy^2}{2} - \frac{y^3}{6}\right]_0^h = \frac{(\rho g\sin\alpha)^2 h^3}{3\mu} \tag{12}$$

Conservation of energy. In the steady total-energy balance of V_o (Notes.pdf, Eq. 4.3 for a fixed volume), each surface contributes as follows:

Surface	Convective flux	Pressure work	Viscous work	Heat
Wall Σ_w	0 ($\bar{v}\cdot\bar{n}=0$)	0	0 ($\bar{v}=0$)	$-(\bar{q}\cdot\bar{n})_w$
Free surface Σ_s	0 ($v_y=0$)	0 ($v_y=0$)	0 ($\bar{\bar{\tau}}'\cdot\bar{n}=0$)	0 (adiabatic)
Faces normal to x	cancel (profiles independent of x)	cancel	cancel	0

The only remaining term besides the heat flux is the work of the mass force, so $0=\int_{V_o}\rho\bar{g}\cdot\bar{v}\,dV-(\bar{q}\cdot\bar{n})_w$. The internal-energy balance (Notes.pdf, Eq. 4.25), $0=\int_{V_o}\phi_v\,dV-(\bar{q}\cdot\bar{n})_w$ with no compression work, gives the second equality:

$$\boxed{(\bar{q}\cdot\bar{n})_w=\int_{V_o}\phi_v\,dV=\int_{V_o}\rho\bar{v}\cdot\bar{g}\,dV=\frac{(\rho g\sin\alpha)^2h^3}{3\mu}}\qquad(13)$$

Equations (10), (11) and (12) confirm it. The work of gravity on the liquid is entirely dissipated by viscosity and leaves through the wall as heat; the kinetic energy of the liquid in V_o does not change.

PHASE 4

Interpretation, Limits and Dimensional Check

- **Physical meaning.** The profile (6) is half a parabola: the weight component along the slope is balanced by the shear on the wall, $\tau_w=\rho gh\sin\alpha$, and the pressure is hydrostatic with the normal component of gravity. The wall carries the weight component normal to the plane as pressure.
- **Limits.** For $\alpha=0$ the plane is horizontal and nothing drives the flow: $v_x=0$, no dissipation, no heat flux. For $\alpha=90^\circ$ (vertical wall) the pressure is uniform and equal to p_a and the shear equals the weight of the layer per unit area. The mean velocity is $\bar{v}=gh^2\sin\alpha/(3\nu)$ (flux $Q=\int_0^h v_x\,dy=gh^3\sin\alpha/(3\nu)$).
- **Dimensional check.** gh^2/ν has units $(\mathrm{m/s^2})\,\mathrm{m^2/(m^2/s)}=\mathrm{m/s}$. $\rho gh\sin\alpha$ is a pressure. $(\rho g)^2h^3/\mu$ has units $(\mathrm{N/m^3})^2\,\mathrm{m^3/(Pa\,s)}=\mathrm{N^2\,m^{-3}/(N\,m^{-2}\,s)}=\mathrm{N/(m\,s)}=\mathrm{W/m^2}$. In (9), $(\rho g)^2h^4/(\mu k)$ has units K.

- **Frame check.** With $\bar{e}_X$ horizontal (towards the descending side) and $\bar{e}_Z$ vertical upward, $\bar{e}_x = \cos\alpha\,\bar{e}_X - \sin\alpha\,\bar{e}_Z$ and $\bar{e}_y = \sin\alpha\,\bar{e}_X + \cos\alpha\,\bar{e}_Z$. The matrix of components has $\det R = \cos^2\alpha + \sin^2\alpha = 1$ and $R\,R^T = I$. Then $\bar{g}\cdot\bar{e}_x = (-g\,\bar{e}_Z)\cdot\bar{e}_x = g\sin\alpha$ and $\bar{g}\cdot\bar{e}_y = -g\cos\alpha$, the components given in the statement.